

Experiment 1: Rogue Many

This is an extension of [Experiment 1: Rogue One](#) where instead of a single rogue point, we have multiple rogue points in a cluster with the same label and the rogue point from the previous experiment as the centroid of the cluster.

[!NOTE]

The Amnesiac unlearning method does not have a repair step. Might not be necessary for this experiment as we are forgetting few points and points are easily separable.

[!NOTE]

The SAE model is slightly different from the other models (it uses skip connections) and the results before unlearning will be different than the others, even when using the same seed. The same is the case for the SISA model as it is an ensemble of models.

How to use

To generate the data, run:

```
python exp1_rogue_many/generate_data.py
```

To run the experiment for all the datasets and unlearning methods, run:

```
python exp1_rogue_many/experiment_run.py
data.dataset="data/data_1.npz,data/data_2.npz,data/data_3.npz,data/data_4.
npz,data/data_5.npz" unlearn.method="amnesiac,ssd,retrain,scrubr,sae,sisa"
hydra/launcher=ray --multirun
```

[!TIP] The `hydra/launcher=ray` argument will run the experiments in parallel using Ray.

You can overwrite any parameters from the config file, for example `n_epochs`:

```
python exp1_rogue_many/experiment_run.py trainer.n_epochs=10
```

The network

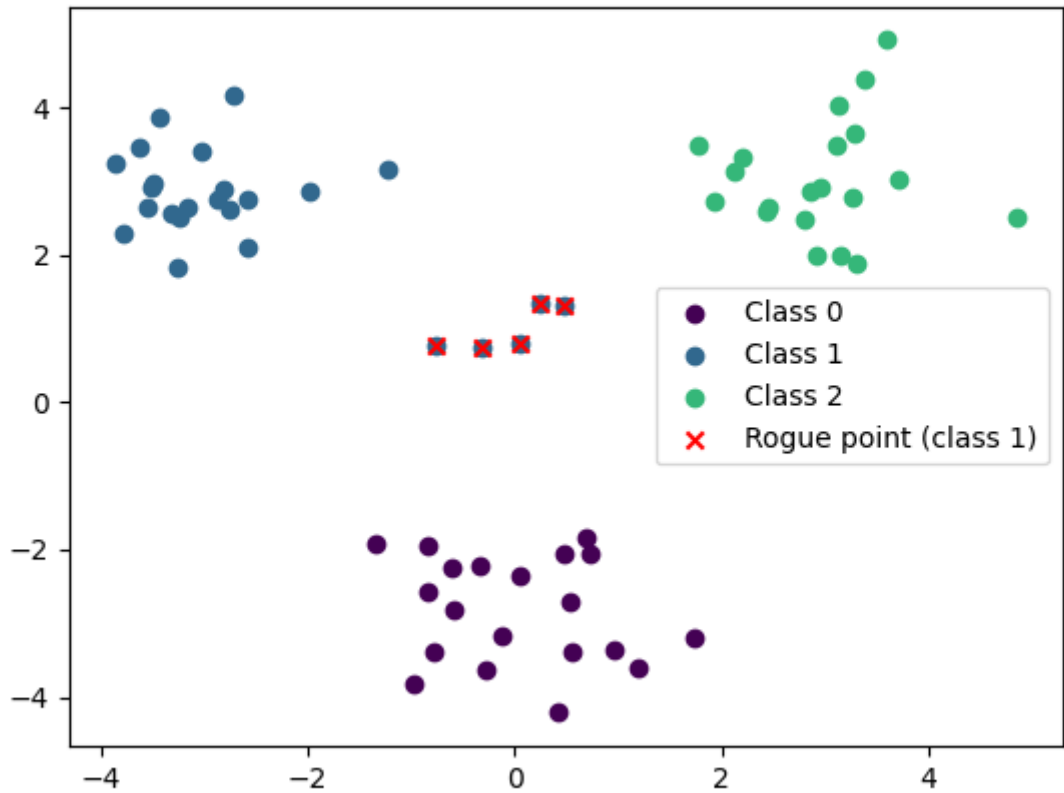
The network is a simple feedforward network which can be seen below:



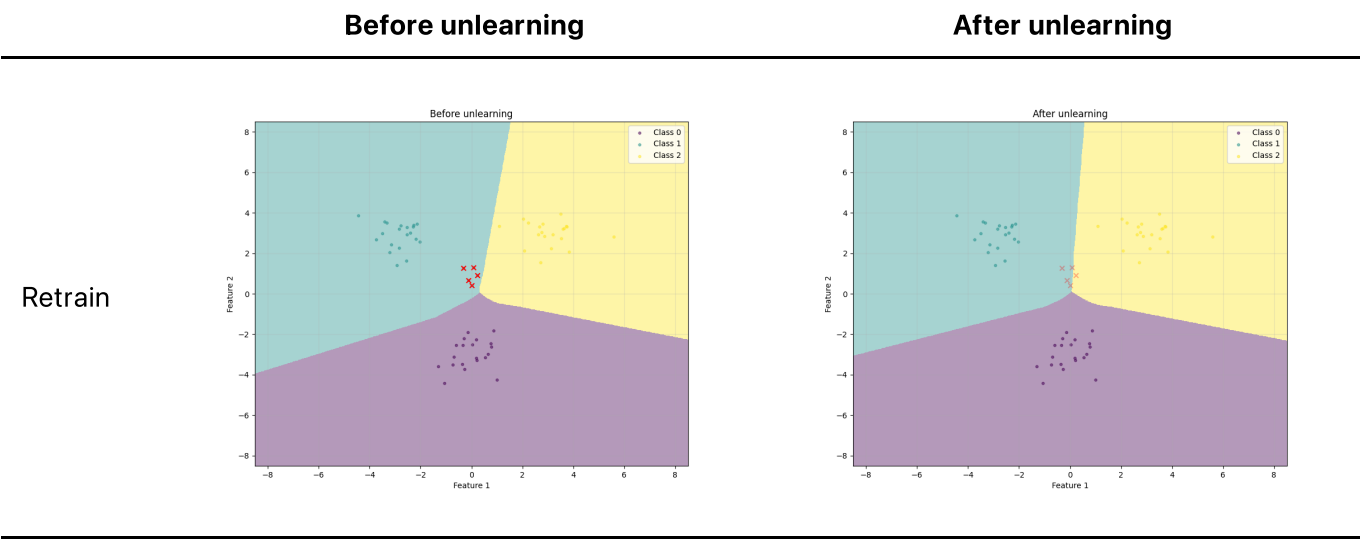
Experiment runs

Now we run the experiment on each of the dataset configuration, moving the rogue point.

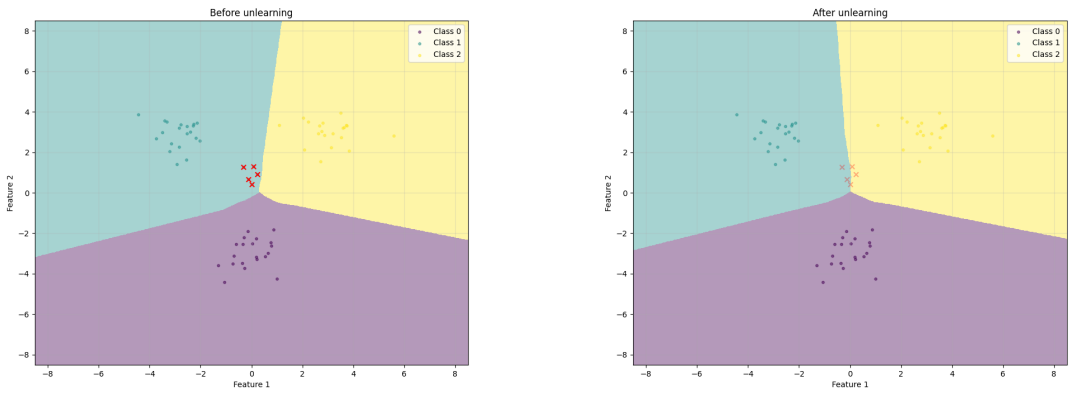
1. Rogue point with same distance to all centroids (close to decision boundary)



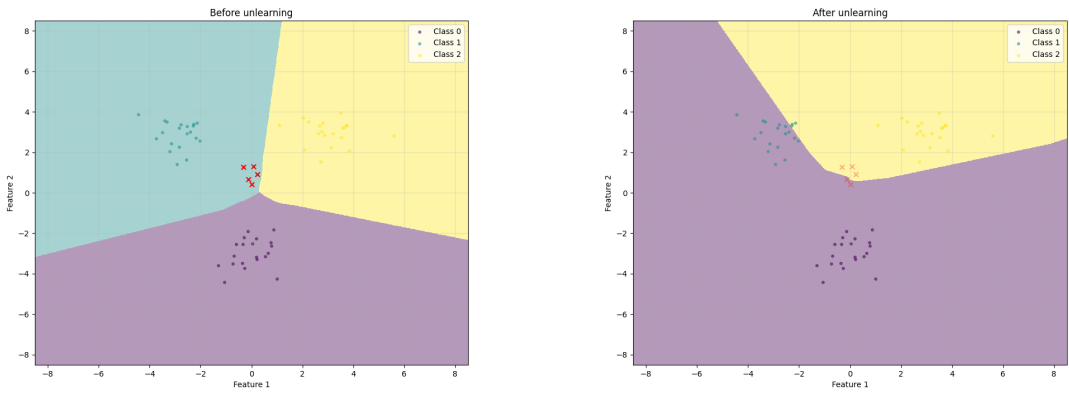
Findings



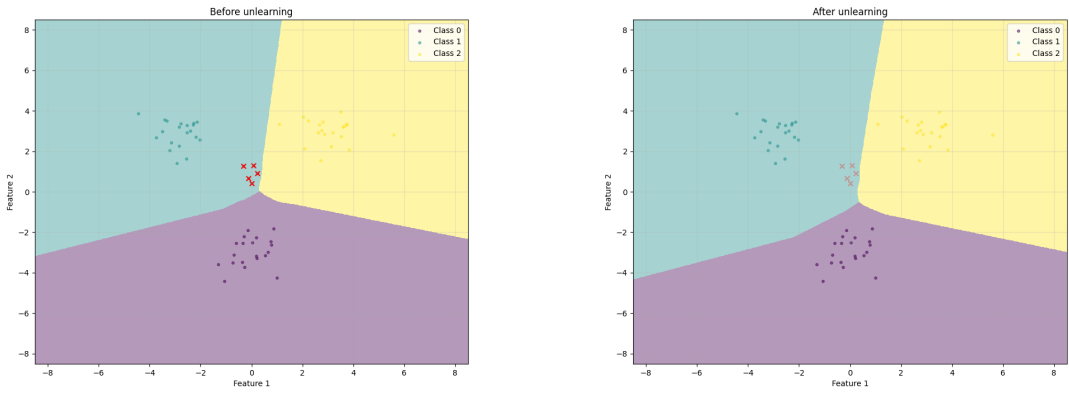
Scrub+R



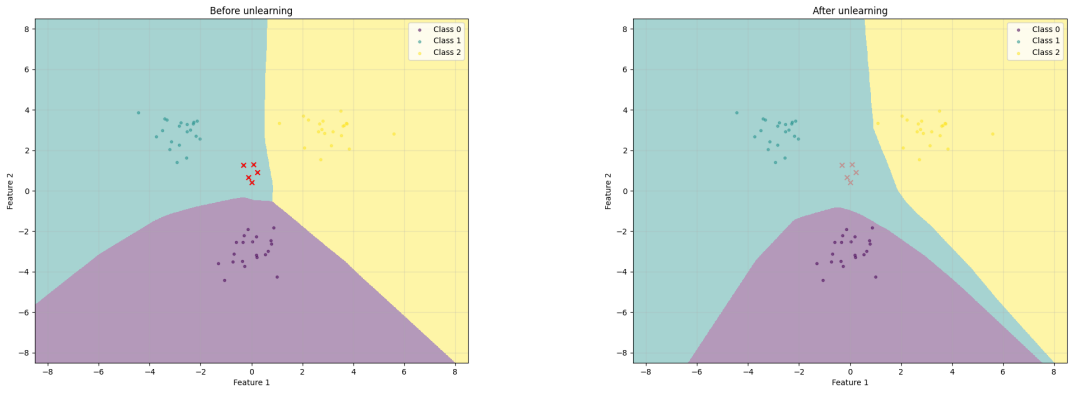
SSD



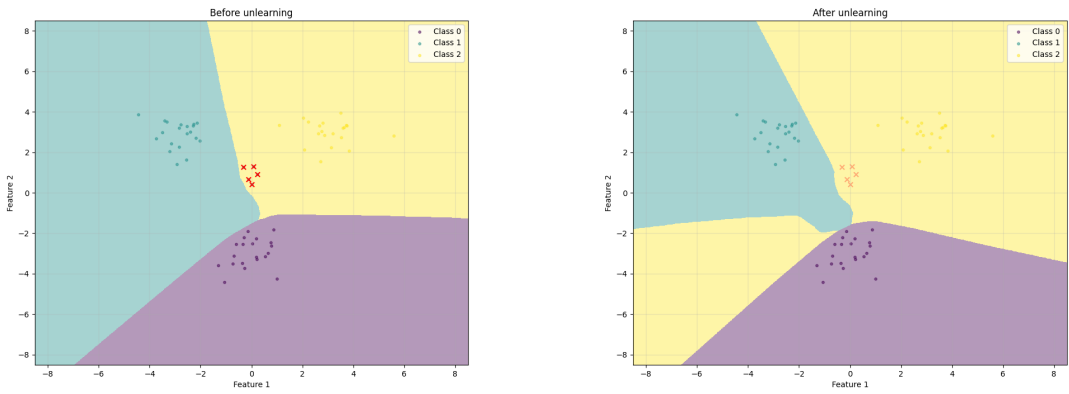
Amnesiac



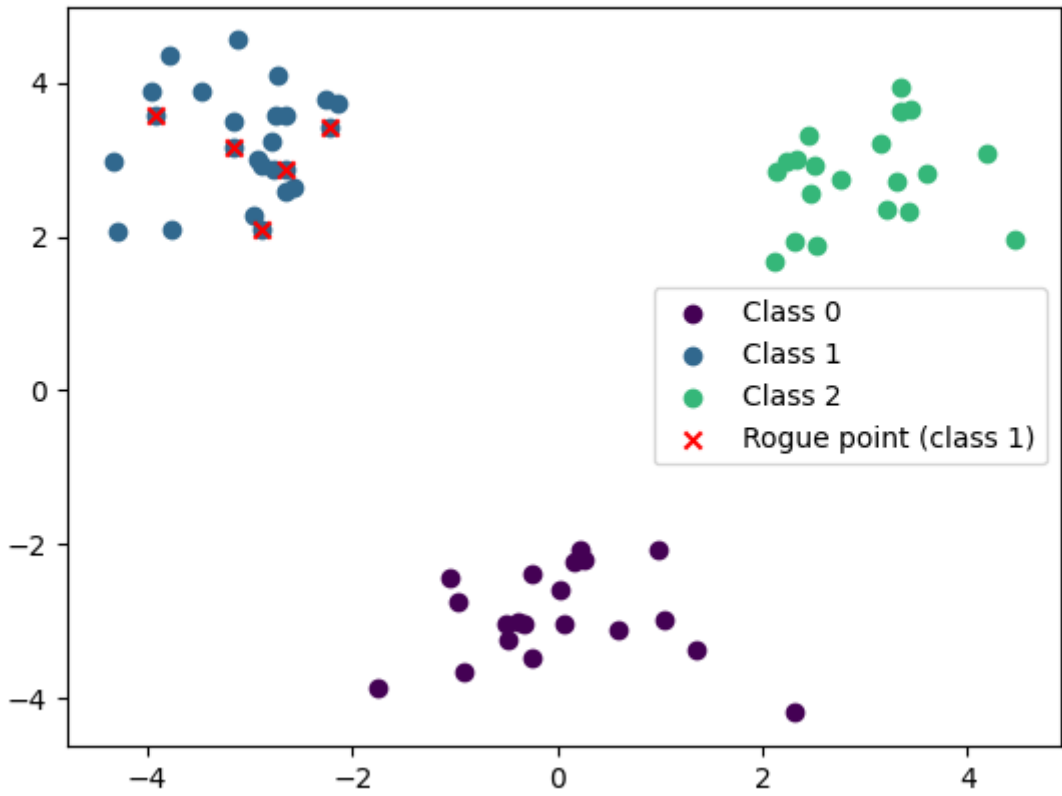
SAE



SISA

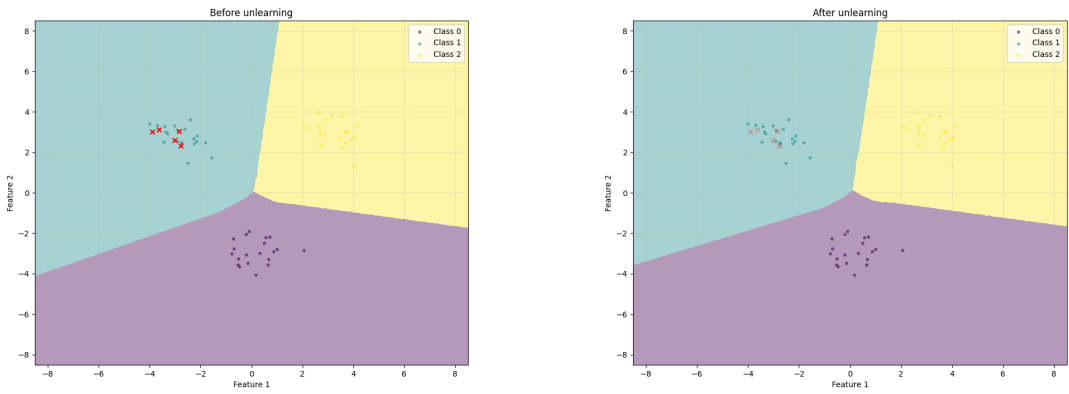


2. Rogue point with same centroid as its class

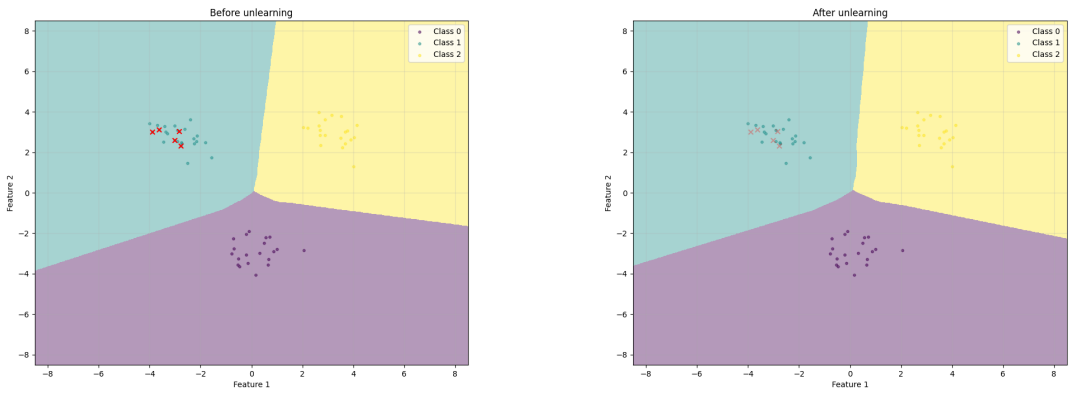


Findings

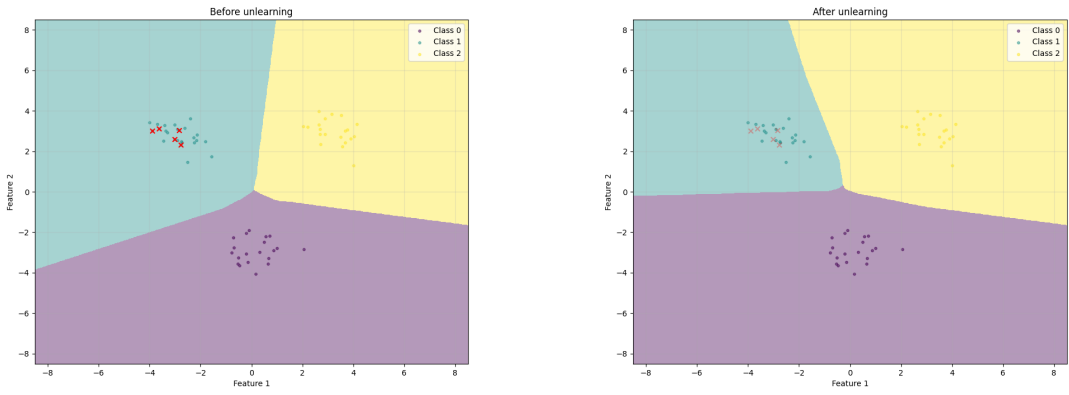
	Before unlearning	After unlearning
Retrain		



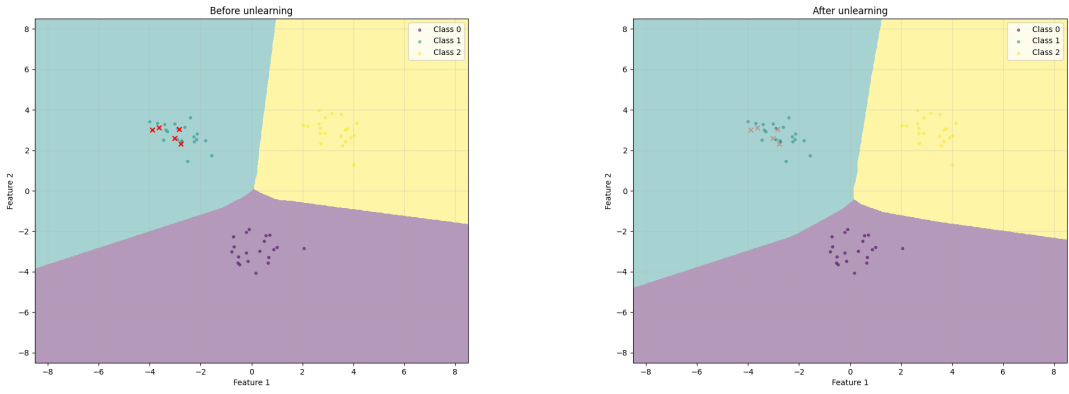
Scrub+R



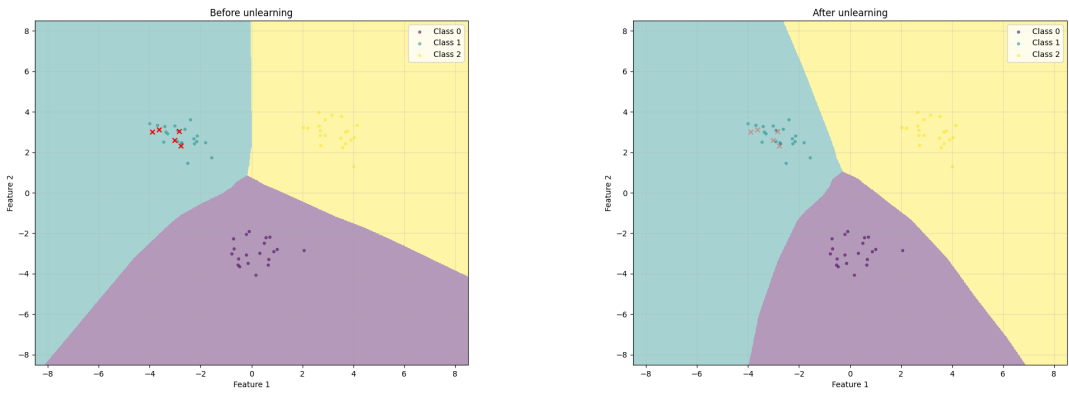
SSD



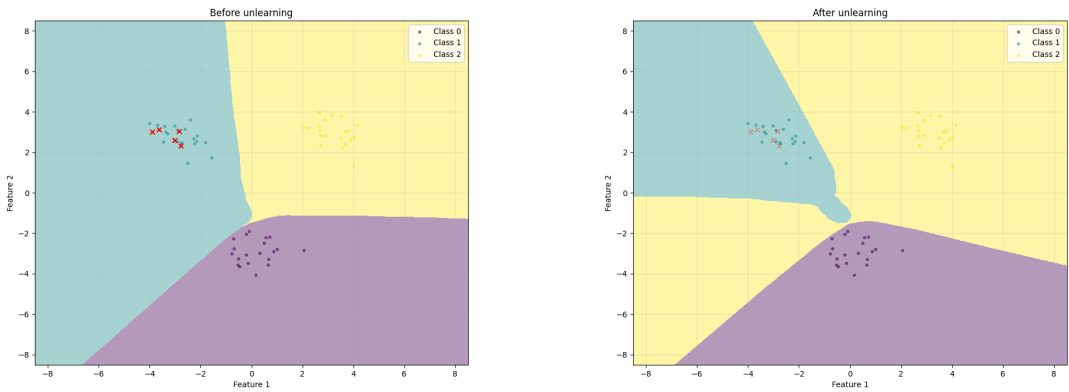
Amnesiac



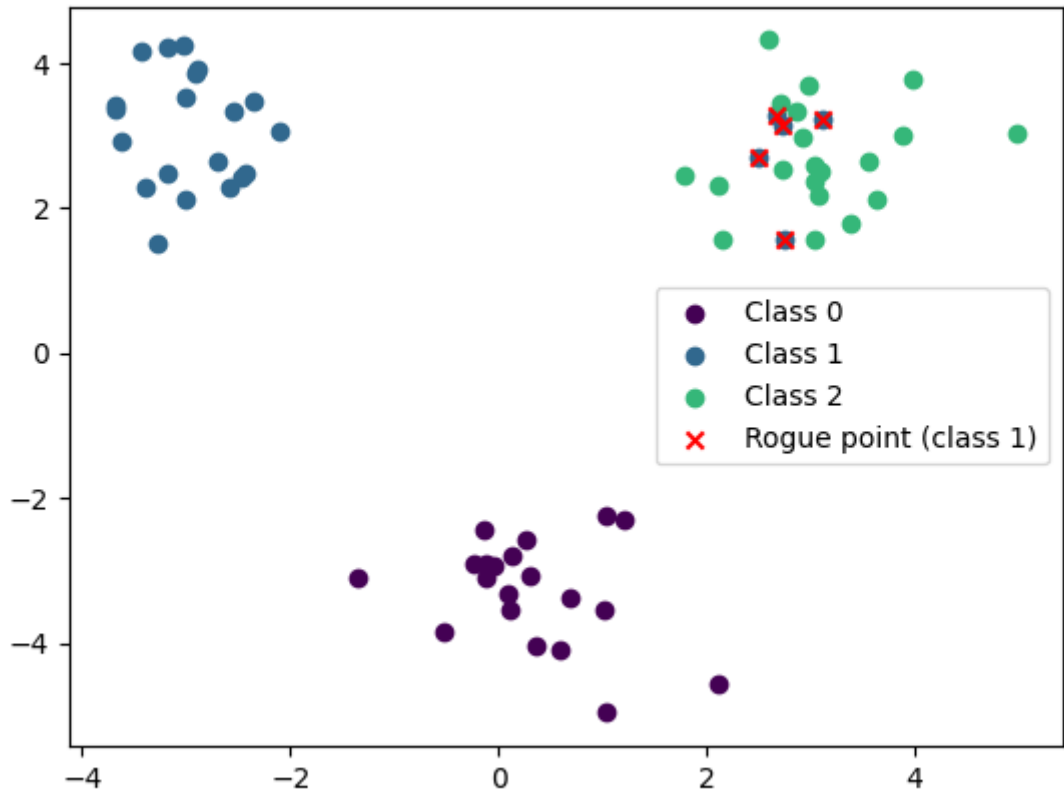
SAE



SISA



3. Rogue point with same centroid as the class with a different label

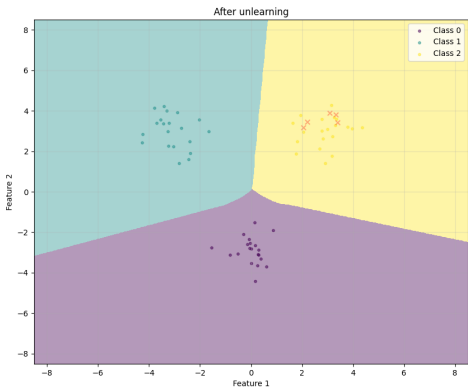
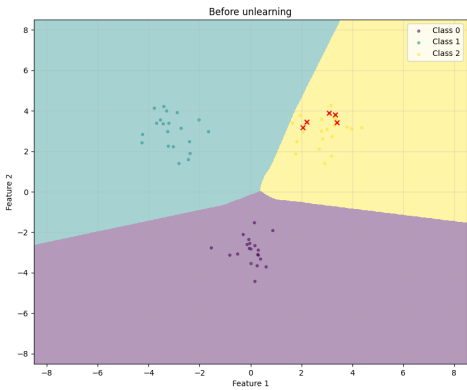


Findings

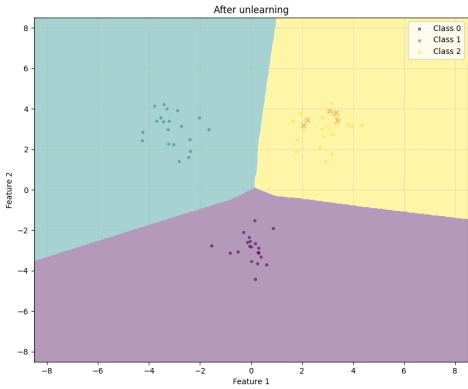
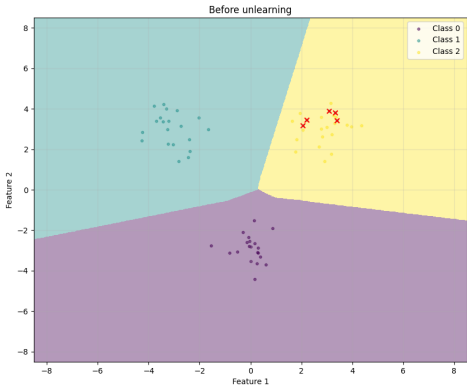
Before unlearning

After unlearning

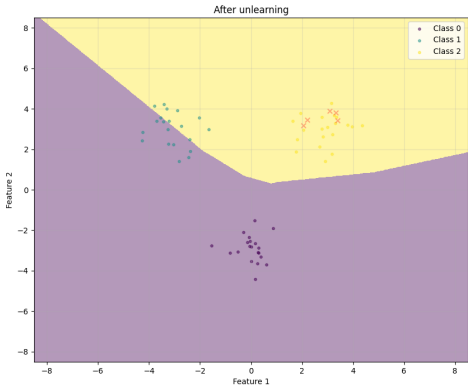
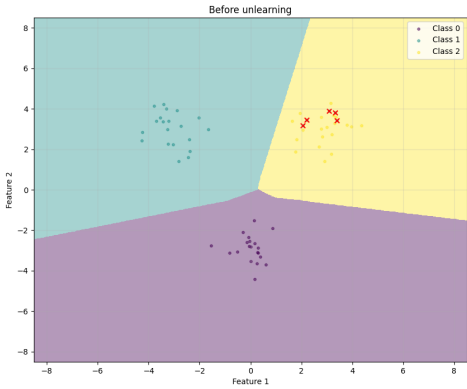
Retrain



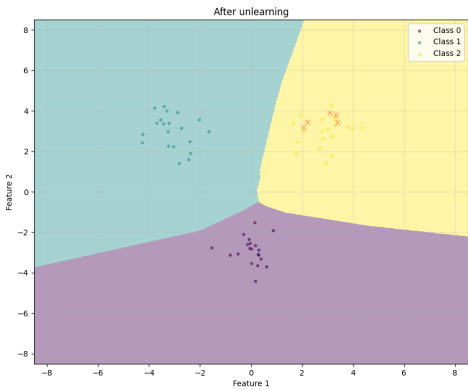
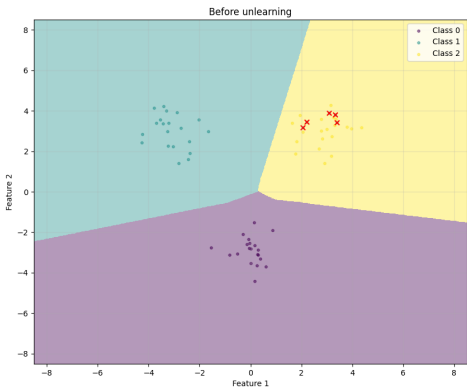
Scrub+R



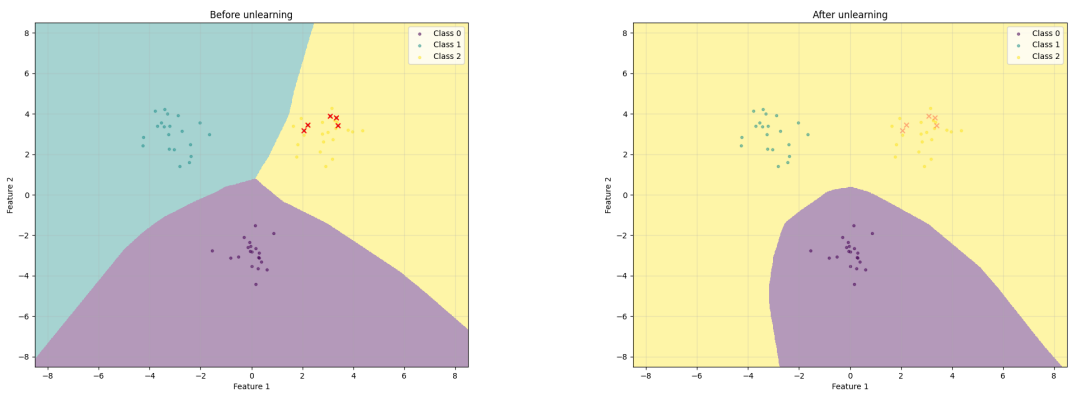
SSD



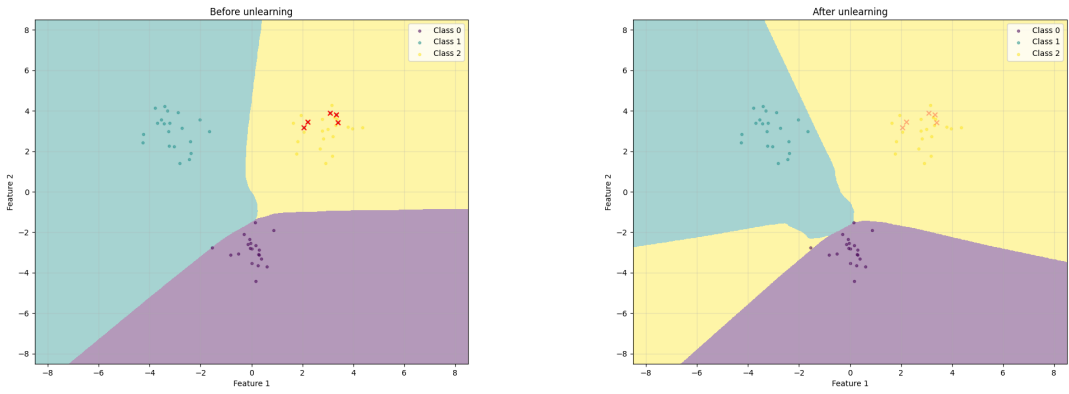
Amnesiac



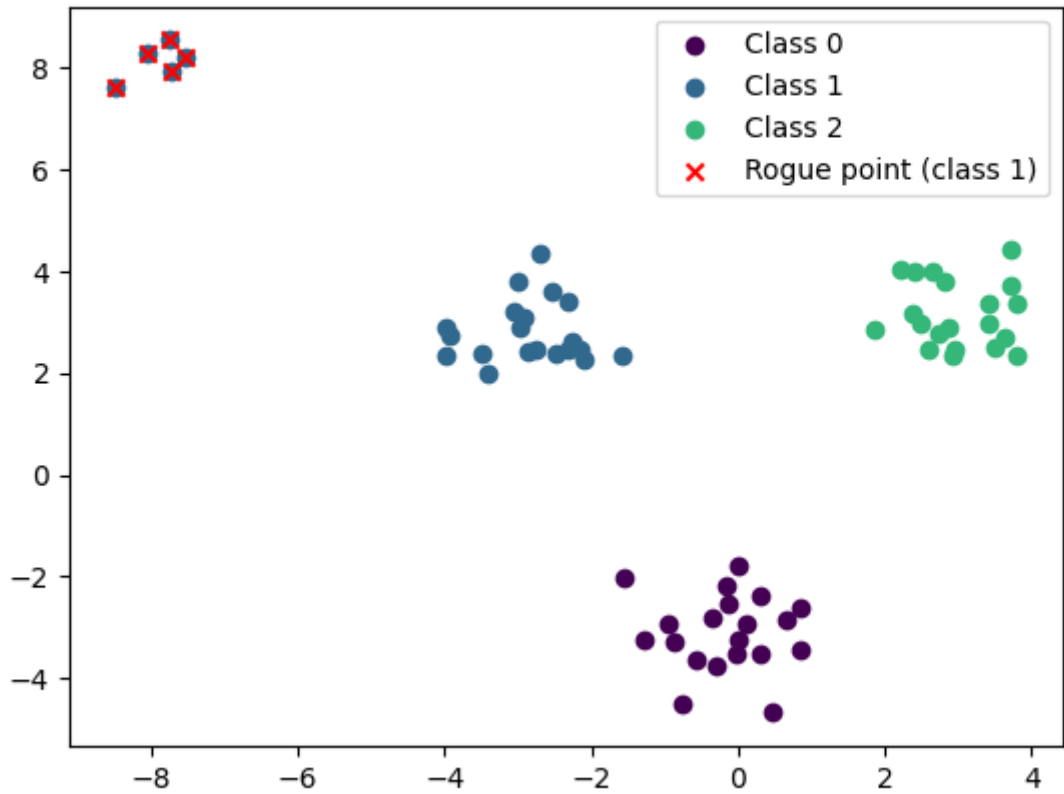
SAE



SISA



4. Rogue point far away from its centroid, but probably in the same decision boundary

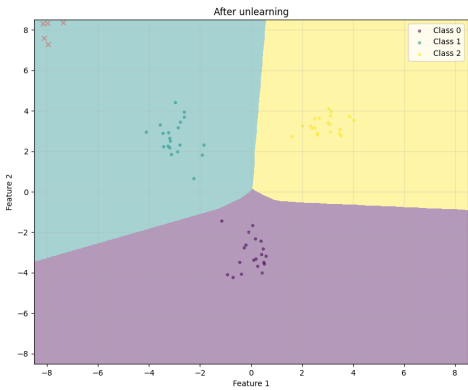
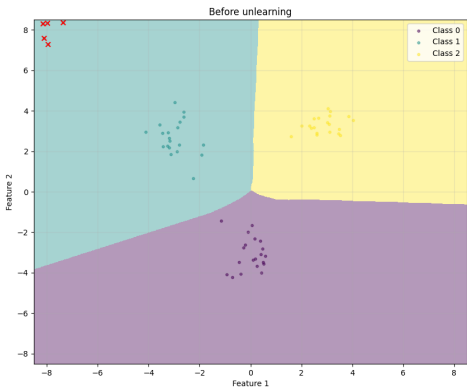


Findings

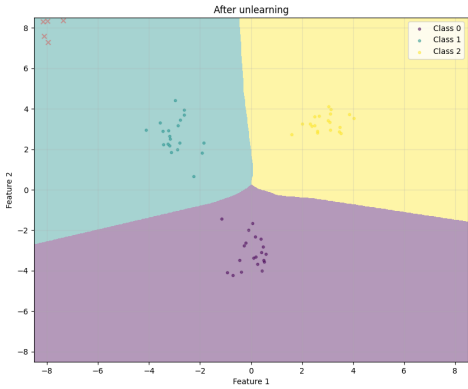
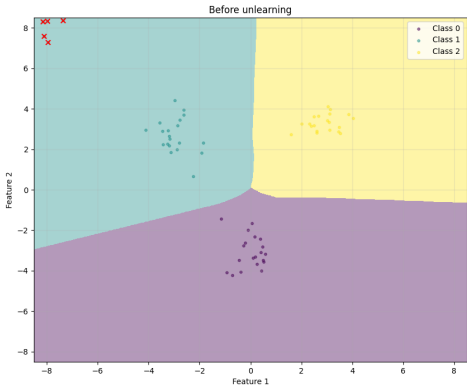
Before unlearning

After unlearning

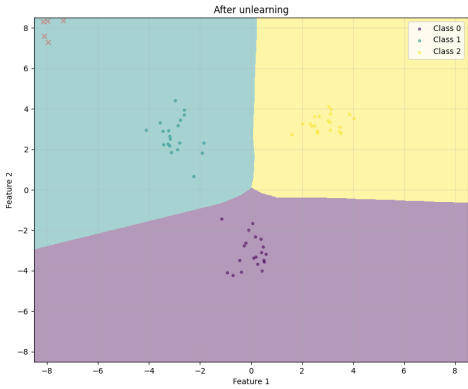
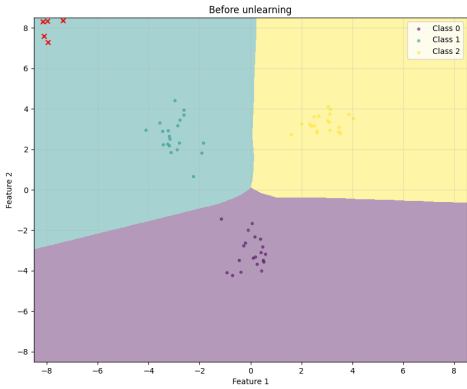
Retrain



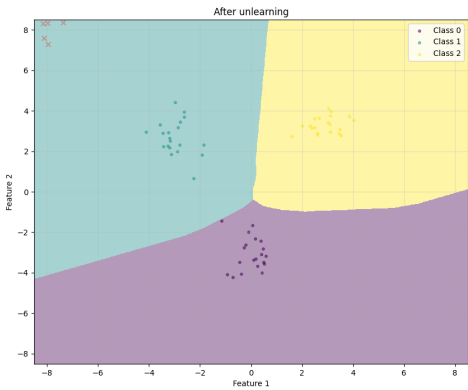
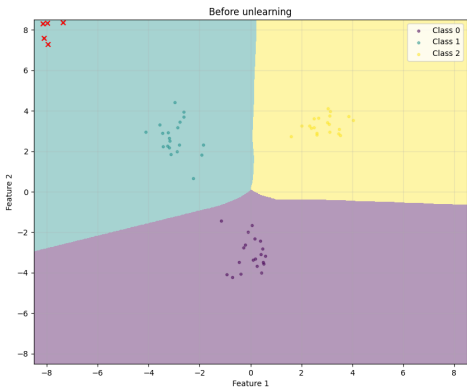
Scrub+R



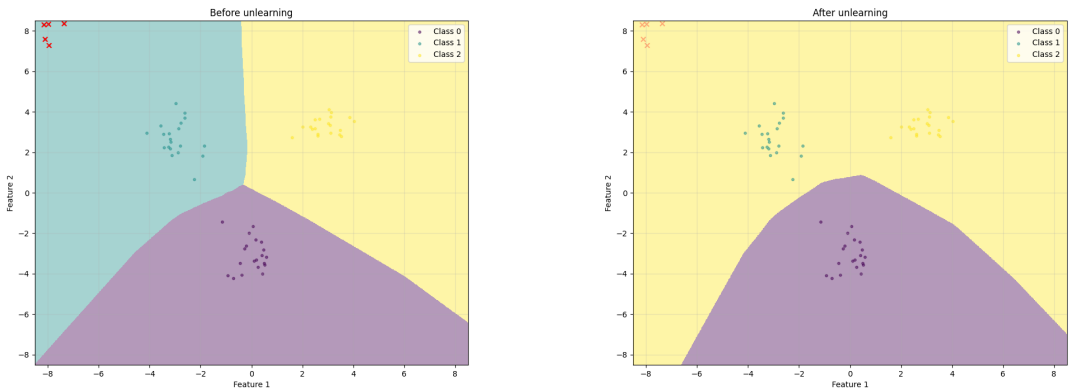
SSD



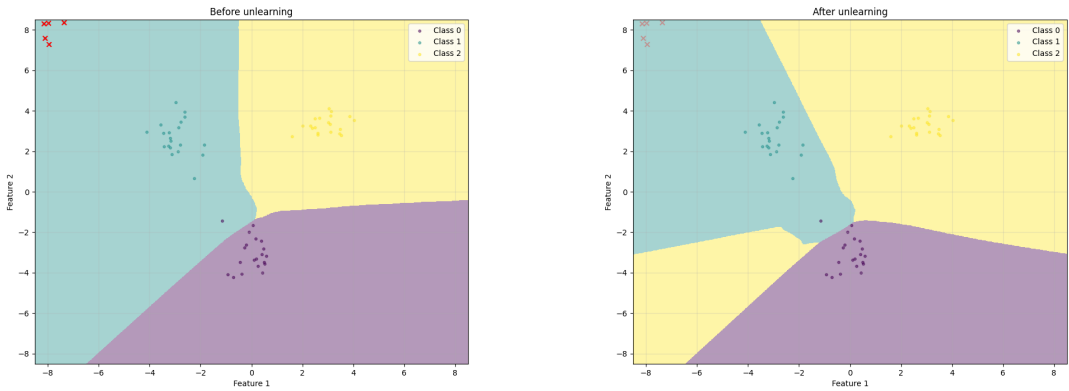
Amnesiac



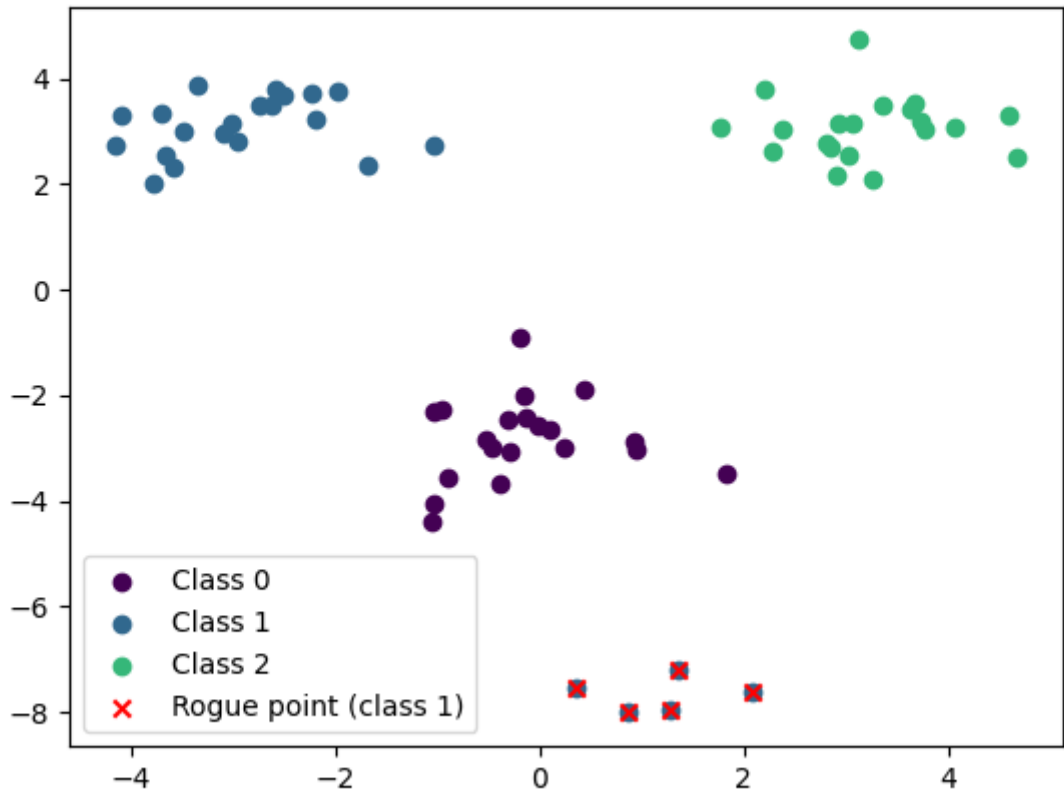
SAE



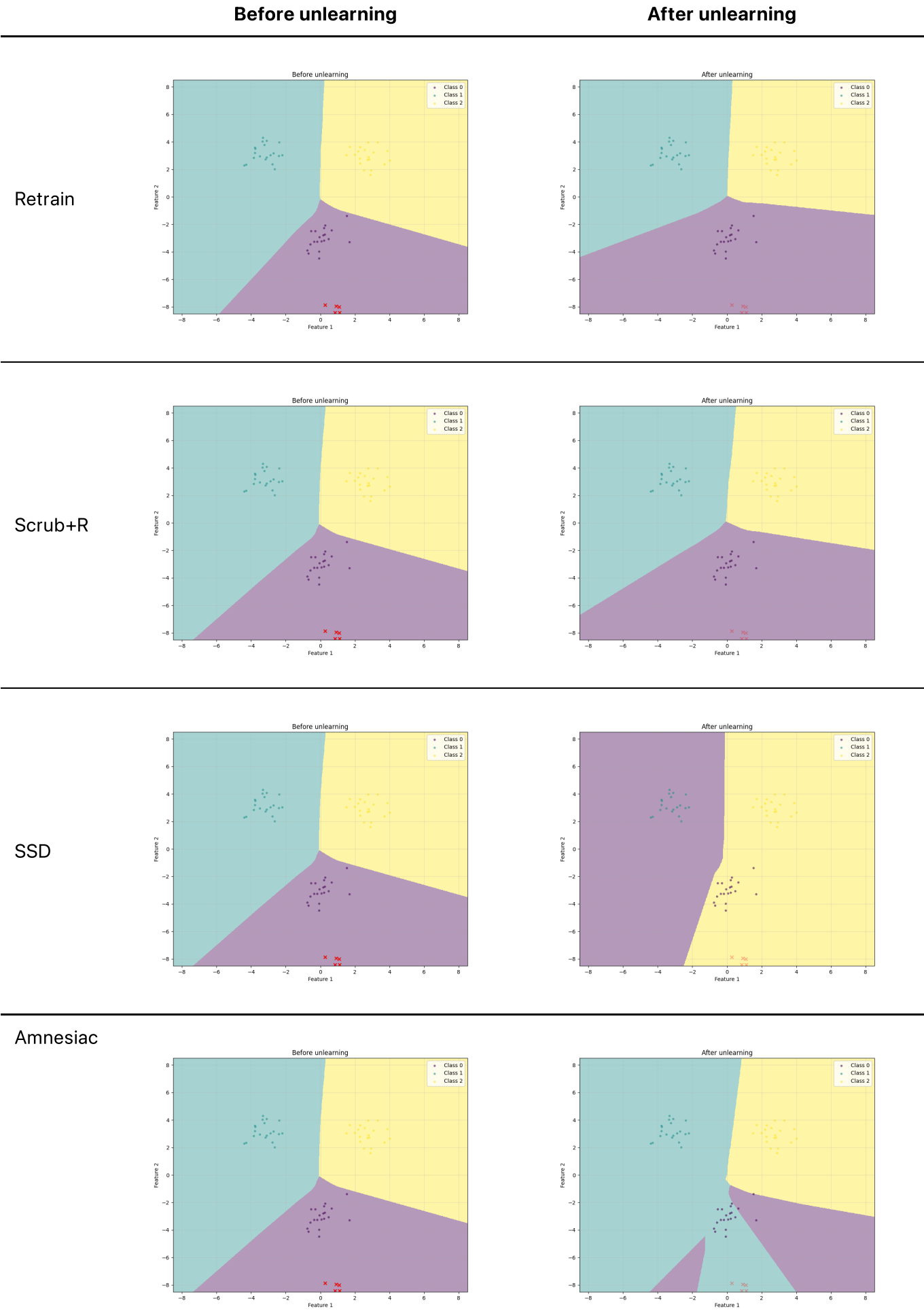
SISA



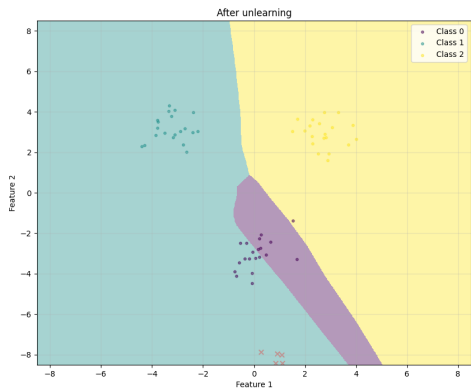
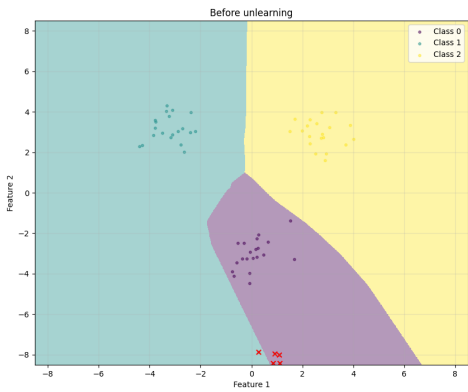
5. Rogue point far away from its centroid, but probably in the same decision boundary



Findings



SAE



SISA

